

VALOXTM FR RESIN ENH4530

REGION ASIA

DESCRIPTION

VALOX ENH4530 is a 15% glass reinforced, non-chlorinated/brominated flame retardant Polybutylene Terephthalate (PBT) injection moldable grade with excellent chemical resistance. It has a UL94V0@0.80mm flame rating. This is a good candidate for a variety of applications needing a sustainable FR PBT solution.

TYPICAL PROPERTY VALUES

Revision 20190925

PROPERTIES	TYPICAL VALUES	UNITS	TEST METHODS
MECHANICAL			
Tensile Stress, yld, Type I, 5 mm/min	80	MPa	ASTM D 638
Tensile Stress, brk, Type I, 5 mm/min	80	MPa	ASTM D 638
Tensile Strain, yld, Type I, 5 mm/min	2	%	ASTM D 638
Tensile Strain, brk, Type I, 5 mm/min	3	%	ASTM D 638
Tensile Modulus, 5 mm/min	7300	MPa	ASTM D 638
Flexural Stress, yld, 1.3 mm/min, 50 mm span	120	MPa	ASTM D 790
Flexural Stress, brk, 1.3 mm/min, 50 mm span	120	MPa	ASTM D 790
Flexural Modulus, 1.3 mm/min, 50 mm span	5800	MPa	ASTM D 790
Tensile Stress, yield, 5 mm/min	80	MPa	ISO 527
Tensile Stress, break, 5 mm/min	80	MPa	ISO 527
Tensile Strain, yield, 5 mm/min	2	%	ISO 527
Tensile Strain, break, 5 mm/min	3	%	ISO 527
Tensile Modulus, 1 mm/min	7500	MPa	ISO 527
Flexural Stress, yield, 2 mm/min	130	MPa	ISO 178
Flexural Stress, break, 2 mm/min	130	MPa	ISO 178
Flexural Strain, break, 2 mm/min	3	%	ISO 178
Flexural Modulus, 2 mm/min	6400	MPa	ISO 178
Ball Indentation Hardness, H358/30	128	MPa	ISO 2039-1
Hardness, Rockwell R	98	-	ISO 2039-2
IMPACT			
Charpy Impact, unnotched, 23°C	35	kJ/m ²	ISO 179/2C
Charpy Impact, unnotched, -30°C	25	kJ/m ²	ISO 179/2C
Izod Impact, unnotched, 23°C	400	J/m	ASTM D 4812
Izod Impact, unnotched, -30°C	360	J/m	ASTM D 4812
Izod Impact, notched, 23°C	50	J/m	ASTM D 256
Izod Impact, notched, 0°C	50	J/m	ASTM D 256
Izod Impact, notched, -30°C	50	J/m	ASTM D 256
Izod Impact, unnotched 80*10*4 +23°C	30	kJ/m ²	ISO 180/1U
Izod Impact, unnotched 80*10*4 -30°C	23	kJ/m ²	ISO 180/1U
Izod Impact, notched 80*10*4 +23°C	6	kJ/m ²	ISO 180/1A
Izod Impact, notched 80*10*4 0°C	6	kJ/m ²	ISO 180/1A
Izod Impact, notched 80*10*4 -30°C	6	kJ/m ²	ISO 180/1A
Charpy Impact, notched, 23°C	6	kJ/m ²	ISO 179/2C

PROPERTIES	TYPICAL VALUES	UNITS	TEST METHODS
Charpy Impact, notched, -30°C	6	kJ/m ²	ISO 179/2C
Charpy 23°C, V-notch Edgew 80*10*4 sp=62mm	4	kJ/m ²	ISO 179/1eA
THERMAL			
CTE, -40°C to 95°C, flow	3.50E-05	1/°C	ASTM E 831
CTE, -40°C to 95°C, xflow	7.00E-05	1/°C	ASTM E 831
Vicat Softening Temp, Rate A/50	217	°C	ASTM D 1525
Vicat Softening Temp, Rate B/50	205	°C	ASTM D 1525
HDT, 0.45 MPa, 3.2 mm, unannealed	220	°C	ASTM D 648
HDT, 1.82 MPa, 3.2mm, unannealed	200	°C	ASTM D 648
CTE, -40°C to 40°C, flow	2.85E-05	1/°C	ISO 11359-2
CTE, -40°C to 40°C, xflow	7.E-05	1/°C	ISO 11359-2
CTE, 23°C to 150°C, flow	4.2E-05	1/°C	ISO 11359-2
CTE, 23°C to 150°C, xflow	1.73E-04	1/°C	ISO 11359-2
Ball Pressure Test, 125°C +/- 2°C	PASSES	-	IEC 60695-10-2
Vicat Softening Temp, Rate A/50	220	°C	ISO 306
Vicat Softening Temp, Rate B/50	205	°C	ISO 306
Vicat Softening Temp, Rate B/120	205	°C	ISO 306
HDT/Bf, 0.45 MPa Flatw 80*10*4 sp=64mm	220	°C	ISO 75/Bf
HDT/Af, 1.8 MPa Flatw 80*10*4 sp=64mm	200	°C	ISO 75/Af
Relative Temp Index, Elec ⁽¹⁾	140	°C	UL 746B
Relative Temp Index, Mech w/impact ⁽¹⁾	120	°C	UL 746B
Relative Temp Index, Mech w/o impact ⁽¹⁾	140	°C	UL 746B
PHYSICAL			
Specific Gravity	1.45	-	ASTM D 792
Filler Content	15	%	ASTM D 229
Moisture Absorption, 50% RH, 24 hrs	0.07	%	ASTM D 570
Mold Shrinkage on Tensile Bar, flow	0.3 – 0.8	%	SABIC method
Melt Flow Rate, 265°C/5.0 kgf	24	g/10 min	ASTM D 1238
Density	1.45	g/cm ³	ISO 1183
Water Absorption, (23°C/sat)	0.15	%	ISO 62
Moisture Absorption (23°C / 50% RH)	0.07	%	ISO 62
Melt Volume Rate, MVR at 250°C/5.0 kg	17	cm ³ /10 min	ISO 1133
Melt Volume Rate, MVR at 265°C/5.0 kg	18	cm ³ /10 min	ISO 1133
Melt Viscosity, 260°C, 1500 sec-1	150	Pa-s	ISO 11443
ELECTRICAL			
Volume Resistivity	1.E+15	Ohm-cm	ASTM D 257
Dielectric Strength, in oil, 0.8 mm	28	kV/mm	ASTM D 149
Dielectric Strength, in oil, 1.6 mm	21	kV/mm	ASTM D 149
Dielectric Strength, in oil, 3.2 mm	17	kV/mm	ASTM D 149
Relative Permittivity, 1 MHz	3.7	-	ASTM D 150
Dissipation Factor, 1 MHz	0.15	-	ASTM D 150
Volume Resistivity	1.E+15	Ohm-cm	IEC 60093
Surface Resistivity, ROA	>1.E+15	Ohm	IEC 60093
Dielectric Strength, in oil, 0.8 mm	28	kV/mm	IEC 60243-1
Dielectric Strength, in oil, 1.6 mm	21	kV/mm	IEC 60243-1

PROPERTIES	TYPICAL VALUES	UNITS	TEST METHODS
Dielectric Strength, in oil, 3.2 mm	17	kV/mm	IEC 60243-1
Relative Permittivity, 1 MHz	3.7	-	IEC 60250
Dissipation Factor, 1 MHz	0.15	-	IEC 60250
Comparative Tracking Index (UL) {PLC}	1	PLC Code	UL 746A
Hot-Wire Ignition (HWI), PLC 0	≥0.8	mm	UL 746A
High Amp Arc Ignition (HAI), PLC 0	≥0.8	mm	UL 746A
FLAME CHARACTERISTICS ⁽¹⁾			
UL Yellow Card Link	E207780-100897495	-	-
UL Recognized, 94-5VA Flame Class Rating	≥2	mm	UL 94
UL Recognized, 94V-0 Flame Class Rating	≥0.5	mm	UL 94
Glow Wire Ignitability Temperature, 3.0 mm	750	°C	IEC 60695-2-13
Glow Wire Ignitability Temperature, 1.5 mm	750	°C	IEC 60695-2-13
Glow Wire Ignitability Temperature, 0.8 mm	750	°C	IEC 60695-2-13
Glow Wire Flammability Index, 3.0 mm	960	°C	IEC 60695-2-12
Glow Wire Flammability Index, 1.5 mm	960	°C	IEC 60695-2-12
Glow Wire Flammability Index, 0.8 mm	960	°C	IEC 60695-2-12
Oxygen Index (LOI)	31	%	ASTM D 2863
Oxygen Index (LOI)	31	%	ISO 4589
INJECTION MOLDING			
Drying Temperature	110 – 120	°C	
Drying Time	2 – 4	hrs	
Maximum Moisture Content	0.02	%	
Melt Temperature	245 – 260	°C	
Nozzle Temperature	230 – 255	°C	
Front - Zone 3 Temperature	240 – 260	°C	
Middle - Zone 2 Temperature	235 – 250	°C	
Rear - Zone 1 Temperature	230 – 240	°C	
Hopper Temperature	40 – 60	°C	
Mold Temperature	40 – 100	°C	

(1) UL Ratings shown on the technical datasheet might not cover the full range of thicknesses and colors. For details, please see the UL Yellow Card.

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